

PAPER_S205 — Session 205 Backfill: Phase H205 $E^\pm(t)$ Expansion–Erosion Footprint (Universal Buoyancy Counter-Balance, $F_{U,Bi}/F_U$)

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Abstract

Session 205 introduced three standalone modules: `positive_et_expansion.py` (positive-energy expansion engine $E^+(t)$, Kozima neutron-drop coupling, expansion Lagrangian), `negative_et_erosion.py` (negative-energy erosion engine $E^-(t)$, net-energy balance, gravitational-wave damping, erosion Lagrangian), and `uqff_vs_string_comparison.py` (10-aspect weighted UQFF vs. String-Theory comparison). The master equations encode the universal-buoyancy counter-balance $F_{U,Bi}/F_U$ as the single algebraic parameter that governs whether vacuum energy expands, erodes, or balances. This paper records the six exact identities (the H205 closures) that pin that algebra in the closure ledger. Three of the six are closed-form calculus / algebra theorems (H205-1 doubling-time, H205-2 critical balance, H205-3 net-factor); two are definitional arithmetic checkpoints (H205-4 weight-sum, H205-6 test count); one is the postulated VDS dimensionality (H205-5, $N_{\text{levels}} = 26$). No claim is made on the overall winner of the UQFF vs. String comparison; the closures only verify that the comparison’s scoring weights are a probability simplex.

1 Ledger Summary

ID	Closure	Value	Class
H205-1	Doubling-time identity $t_{\text{double}}(\kappa + [\text{SSq}]/26) = \ln 2$	31.6172 s	THEOREM
H205-2	Critical balance ratio r_{crit} solving $2r - 1 = 0$	1/2	THEOREM
H205-3	Net-factor formula $\text{nf}(r) = 2r - 1$, $r = 1.1$	1.2	THEOREM
H205-4	Comparison-weight unit sum	1.00	arithmetic
H205-5	VDS spacetime level count N_{levels}	26	CONVENTION
H205-6	S205 module-test total $9 + 10 + 14$	33	arithmetic

All six register as EXACT in `master_closures.csv` / `sigma_table.csv` (script: `_session205_closures.py`; final parseable line S205-doubling-time-identity: 31.6172 vs 31.6172 -> EXACT). Independent verification is provided by `_verify_s205.py`.

2 Theorems — Calculus and Algebraic Identities

Theorem 1 (H205-1 — Doubling-time identity). *Let $E^+(t) = E_0 \exp((\kappa + [\text{SSq}]/N_{\text{lev}})t) \cdot S_{26}([\text{SSq}]) \cdot (F_{U,Bi}/F_U)$, with all factors other than $\exp(\cdot)$ time-independent. Then the doubling time t_{double}*

satisfying $E^+(t_{\text{double}}) = 2 E^+(0)$ is

$$t_{\text{double}} = \frac{\ln 2}{\kappa + [\text{SSq}]/N_{\text{lev}}}.$$

Proof. $E^+(t)/E^+(0) = \exp((\kappa + [\text{SSq}]/N_{\text{lev}})t)$ since the S_{26} and buoyancy-ratio factors cancel. Setting this equal to 2 and taking $\ln$ gives the stated identity. $\square$

With canonical $\kappa = 5 \times 10^{-4} \text{ day}^{-1} = 5.787 \times 10^{-9} \text{ s}^{-1}$, $[\text{SSq}] = 0.57$, $N_{\text{lev}} = 26$:

$$t_{\text{double}} = \frac{\ln 2}{5.787 \times 10^{-9} + 0.57/26} = \frac{0.6931 \dots}{2.1923 \dots \times 10^{-2}} = 31.6172 \dots \text{ s},$$

matching the module test T_5 output **3.1617e+01 s** byte-for-byte.

Theorem 2 (H205-2 — Critical balance ratio). *The net-energy equation*

$$E_{\text{net}}(t) = E_0 \exp((\kappa + [\text{SSq}]/N_{\text{lev}})t) S_{26}([\text{SSq}]) [2(F_{U,Bi}/F_U) - 1]$$

admits the unique zero $E_{\text{net}}(t) \equiv 0$ at $r := F_{U,Bi}/F_U = 1/2$.

Proof. The exponential and S_{26} factors are strictly positive. Hence $E_{\text{net}} = 0$ iff $2r - 1 = 0$, which gives the unique root $r = 1/2$. $\square$

Theorem 3 (H205-3 — Net-factor formula). *Define $\text{nf}(r) := r - (1 - r) = 2r - 1$, where $r = F_{U,Bi}/F_U$. Then for $r = 1.1$, $\text{nf}(1.1) = 1.2$ exactly.*

Proof. $\text{nf}(1.1) = 2(1.1) - 1 = 2.2 - 1 = 1.2$. $\square$

3 Structural Checkpoints — Arithmetic and Convention

Proposition 4 (H205-4 — Weight-sum identity). *The category weights $W = \{w_{\text{test}}, w_{\text{pred}}, w_{\text{found}}, w_{\text{math}}\} = \{0.30, 0.30, 0.20, 0.20\}$ used in `uqff_vs_string_comparison.OverallScoreCalculator` satisfy $\sum_{w \in W} w = 1$, i.e. W is a probability simplex.*

Proof. $0.30 + 0.30 + 0.20 + 0.20 = 1.00$ by direct addition. $\square$

Convention 5 (H205-5 — VDS spacetime level count). The Vacuum-Dimensional-Spacetime (VDS) level count is fixed at $N_{\text{lev}} = 26$ throughout the UQFF framework. This appears in the $S_{26}([\text{SSq}])$ factor of $E^\pm(t)$ and in the per-level rate contribution $[\text{SSq}]/N_{\text{lev}}$. It is a postulated structural constant, not a derived quantity in S205.

Proposition 6 (H205-6 — S205 module-test total). *The three S205 standalone modules carry, respectively, 9, 10, and 14 self-test cases (`positive_et_expansion.py`, `negative_et_erosion.py`, `uqff_vs_string_comparison.py`). Their sum is 33, and all 33 pass clean.*

Proof. $9 + 10 + 14 = 33$ by direct addition; module self-tests verified independently by execution. $\square$

4 Honest Classification

Of the six H205 closures: three are closed-form theorems (1, 2, 3); two are definitional arithmetic checkpoints (4, 6); one is a postulated structural constant (5). No physical-content claim is made; in particular, the $E^\pm(t)$ framework's identification of $F_{U,Bi}/F_U = 1/2$ as the cosmological balance point is a model statement of the UQFF Universal-Buoyancy postulate, not a derivation from first principles. The closures verify only that the model's algebra is internally consistent.

5 Reproducibility

- Closure script: `_session205_closures.py`
- Closure JSON: `_session205_closures.json`
- Tier-13 block: `UQFF_UNIFIED_CLOSURE_DERIVATIONS.py`, records H205-1 through H205-6
- Ledger row: `master_closures.csv` (S205 row 6, status OK)
- Sigma row: `sigma_table.csv` (S205 row 6, bucket EXACT)
- Verifier: `_verify_s205.py` (independent algorithm: `sympy` symbolic root, `fractions.Fraction` rational arithmetic, `mpmath` arbitrary-precision logarithm)
- Source modules: `positive_et_expansion.py`, `negative_et_erosion.py`, `uqff_vs_string_comparison.py`
- Canonical record: `VALIDATION_MASTER_INDEX_2.md`, L462 (v5.62, commit 336d98e7).